

I found the proposal record. Let me now read the attached EOI document to provide a comprehensive review.# Review Brief: A Reproducible Software Platform for Coastal Carbonate Chemistry Ocean Acidification Workflows

Part 1 — Comprehensive Summary

Project Overview:

This project aims to develop an open-source, reproducible software platform that streamlines and standardizes workflows for coastal carbonate chemistry, with a focus on ocean acidification research. The platform is designed for researchers and practitioners working with carbonate chemistry datasets, particularly in data-limited coastal regions such as Ghana. The motivation stems from the current reliance on fragmented spreadsheets, local scripts, and manual data processing, which hinders reproducibility, provenance tracking, and scalability across diverse datasets and research contexts. The proposed solution is to encapsulate existing Python-based workflows into a modular, schema-driven software package that integrates quality control, provenance, and reproducible exports, complemented by a lightweight user interface.

Problem Statement and Gap:

The core problem addressed is the lack of standardized, reproducible workflows for coastal carbonate chemistry, which leads to inconsistent data handling, poor auditability, and limited comparability across studies. Existing practices rely heavily on ad hoc scripts and manual interventions, making it difficult to scale analyses or ensure data integrity. The software aims to fill this gap by providing a structured, auditable, and user-friendly platform that supports robust data management and reproducibility.

Proposed Solution (Architecture):

The solution involves refactoring an existing Python workflow into a modular package, incorporating schema validation, provenance/audit tracking, automated testing, and packaging. The architecture will wrap around established tools such as PyCO2SYS/CO2SYS, adding layers for input handling, quality control, and reproducible exports. The platform will offer both command-line and notebook interfaces, with a lightweight graphical UI for broader accessibility. No high-performance computing or dedicated database infrastructure is required; compute and storage needs are modest.

Scope (In/Out):

In scope: Codebase audit, schema definition, modularization, UI development, automated testing, documentation, packaging, and repository publication.

Explicitly out: Large-scale HPC deployment, complex database integration, and support for non-Python workflows.

Target Users and Deployment Context:

The primary users are researchers in oceanography and environmental sciences, especially those working in coastal and data-limited settings. The software is intended for deployment on standard desktop and laptop environments, with support for both command-line and graphical interfaces. Cloud or HPC deployment is not a focus.

Team Composition and Expertise:

The team includes Prof. Edem Mahu (University of Ghana, co-lead of the Ocean Margins Initiative), with key technical contributions from Maurice Oti Edusei (PhD student, available full-time). The proposal references collaboration with the Ocean Margins Initiative and mentions the involvement of the OBVI program. There is an outstanding question regarding the involvement of Fernanda Lecaros Saavedra (MBARI), who has relevant software engineering expertise but is not explicitly listed as a collaborator.

Timeline, Milestones, and Deliverables:

The project is structured in four phases: (1) codebase audit and schema definition, (2) refactoring into a modular package, (3) testing, documentation, and UI development, and (4) user testing, release preparation, and repository publication. The earliest conceivable start date is June 15, 2026, with the latest beneficial start date being August 1, 2026. The key deliverable is a documented Version 1 software release under the MIT License.

Outcomes, Success Metrics, and Impact Claims:

The stated outcomes include improved reproducibility, auditability, and comparability of carbonate chemistry datasets, with a particular impact on research in coastal Ghana and similar regions. Success will be measured by the release of a documented, tested, and user-validated software package, uptake by the target research community, and the ability to scale analyses across diverse datasets. The project claims to address known pain points in the field and to provide foundational infrastructure for future research and collaboration. ¹

Part 2 — Key Points to Review

1. Technical Approach and Feasibility

- The architecture builds on existing Python tools (PyCO2SYS/CO2SYS) and adds schema-driven input handling, quality control, and reproducible exports. This is a sound approach, leveraging established libraries rather than reinventing core functionality.
- Technology choices (Python, modular packaging, lightweight UI) are justified for the target user base and deployment context.
- No evidence of arbitrary technology selection or unjustified "AI/ML" claims. The use of AI-assisted tools is limited to development support (e.g., ChatGPT for debugging).
- **Red flag:** The proposal does not specify how the UI will be implemented or which frameworks will be used. Details on data flow between components are high-level and could be expanded.
- **Quote:** *"The solution builds a workflow layer around existing tools (e.g., PyCO2SYS/CO2SYS), adding schema-driven input handling, quality control, and reproducible exports via a lightweight UI."* (AI-generated summary) ¹

2. Software Engineering Practices

- The proposal mentions automated testing and documentation as part of the deliverables.
- Version control is not currently in place for the existing codebase, but the plan includes repository publication.
- CI/CD, code review, and packaging are referenced as needs, but the proposal does not detail specific strategies or tools.
- **Gap:** Engineering rigor is assumed but not fully specified. No explicit mention of code review processes or CI/CD pipelines.
- **Quote:** *"Professional Research Software Engineering (RSE) support is required across four phases: ... automated testing, packaging, and a lightweight user interface complementing CLI and notebook use."* (AI-generated)

3. Sustainability and Maintenance

- The proposal states that Maurice Oti Edusei will be responsible for maintenance after the engagement concludes.
- No plan for community building, governance, or broader sustainability is described.
- **Gap:** Long-term sustainability and handoff are not addressed beyond naming a single individual.
- **Quote:** *"After a VISS engagement concludes, who in your group would be responsible for maintaining and extending the software? Maurice Oti Edusei."* 1

4. Data Management and FAIR Principles

- The proposal emphasizes schema-driven input handling and provenance/audit tracking, which align with FAIR principles.
- No explicit mention of data standards, formats, metadata, or licensing for data.
- **Gap:** Details on data licensing, metadata standards, and provenance mechanisms are limited.
- **Quote:** *"The solution builds a workflow layer ... adding schema-driven input handling, quality control, and reproducible exports via a lightweight UI."* (AI-generated summary) 1

5. Open Source and Licensing

- The software will be released under the MIT License.
- No discussion of dependency licenses or potential IP conflicts.
- **Quote:** *"The software will be released under the MIT License."* (AI-generated summary) 1

6. Interoperability and Reuse

- The platform is designed to wrap around existing tools (PyCO2SYS/CO2SYS), suggesting some interoperability.
- No mention of APIs, standards, or integration with broader community tools.
- **Gap:** Potential for siloing if APIs and standards are not prioritized.

7. User Engagement and Adoption

- User testing is included as a project phase.
- No evidence of demand beyond the PI's group or plans for co-design, user training, or broader community engagement.
- **Gap:** User engagement strategy is minimal; no training or outreach plans are described.

8. Team and Effort Allocation

- The team includes a PI and a PhD student, with the latter available full-time.
- The proposal references the OBVI program and the Ocean Margins Initiative but does not clarify the involvement of software engineering experts such as Fernanda Lecaros Saavedra.
- **Gap:** Over-reliance on a single student for production code; unclear if sufficient RSE expertise is available.
- **Quote:** *"Maurice Oti Edusei, PhD student, available to work on this project at any time."* 1

9. Risks and Mitigations

- No explicit risk assessment or mitigation strategies are provided.
- **Gap:** The proposal is silent on technical, personnel, and dependency risks.

10. Specific Questions and Concerns to Raise

- What frameworks and technologies will be used for the UI, and how will it be maintained?
- How will the project ensure long-term sustainability and community uptake beyond the initial team?
- Are there plans for code review, CI/CD, and broader engineering rigor?
- How will data standards, metadata, and provenance be implemented and documented?
- Will Fernanda Lecaros Saavedra or other RSEs be involved, and what is their role?
- What is the plan for user engagement, training, and documentation for non-expert users?
- How will the project handle dependency management and potential IP conflicts?
- What are the key risks (technical, personnel, sustainability), and how will they be mitigated?

11. Strengths Worth Highlighting

- The project addresses a well-defined and significant gap in reproducible research workflows for coastal carbonate chemistry.

- The approach leverages existing, trusted scientific tools and aims for modular, open-source development.
- The team demonstrates domain expertise and a clear understanding of the scientific challenges.
- The proposal includes a phased development plan with explicit deliverables and a commitment to open-source release. 1

Note: This review is based on the available summary and EOI data. The full proposal document may contain additional details not accessible in this brief. Gaps and quotes are cited from the proposal record and AI-generated summary. 2